

Architecture as an Asset Class

Revised body section + recommended appendix structure for Recursive Self-Improvement is a Portfolio Optimization Problem

Editorial note

This document contains two artefacts intended to be merged into the main paper. **Part I** is a rewrite of the body section currently titled “Architecture Search as an Asset Class.” It strengthens the framing, makes the connection back to the working Jacobian g_A explicit, and surfaces the cross-row supermodularity argument that should also be carried forward into the Conclusion. **Part II** is a tactical recommendation for the appendix: which existing material to keep, which to merge, what to cut, and the one new appendix needed to support the rewritten body.

Part I Architecture as an Asset Class (revised body)

In Sections 5.1 and 5.2 we showed that two of the four operational rows — deployment (Q) and sensors / data (X) — are already producing measurable scaling laws denominated in dollars. The third row, architecture search (A), has historically resisted that treatment. Architecture work has been the part of quantitative research that looked least like an asset class: experiments are slow, expensive in human attention, and individual runs carry high outcome variance. Within the framework of Section 3, however, a_t^A is a line in the same allocation problem as the other rows, and the same Jacobian object $g_A := \partial J / \partial a_t^A$ that prices it conceptually is also what the corporation must estimate empirically before architecture spending can be priced against deployment, data, or retraining.

This section makes three points. First, the unit economics of architecture search have changed: the cost of a single competent experiment has collapsed below the variance of a single experiment’s outcome, which is what makes statistical inference about g_A possible at all. Second, we report the first measured campaign within this regime — 929 candidate runs over 11 calendar days, totalling roughly \$300 in tokens — and extract from it both a usable estimate of g_A and a description of the distributional shape (staircase rather than smooth) that any honest pricing of the row must respect. Third, we describe how architecture search interacts *multiplicatively* with every other row in the controller, which is what justifies treating it as the deepest-leverage channel in a_t^{op} rather than as overhead.

The asset-class reframing

Treating architecture search as an asset class requires three conditions to hold simultaneously. The action must be priced in dollars (a marginal unit can be bought). The outcome must be priced in dollars (the marginal unit produces a measurable change in the rest of the controller’s value). And the joint distribution of cost and outcome must be learnable from a sample large enough that the empirical mean of the marginal value is not dominated by sampling noise.

Each of the three conditions has historically failed for architecture research. Compute has always been priced in dollars, but human researcher time was not — supply was thin and quality was idiosyncratic. Outcomes have always been priced in dollars eventually, but the lag between an architectural change and a confidently measurable change in deployed return is long enough that few firms ever closed that loop in their internal accounting. And the experiment count was small: the literature is dotted with papers reporting a

handful of architecture variants, which is enough for narrative but not for posterior estimation.

What changed is the third condition. A research harness composed of a single human investigator paired with an agentic experimentation loop produces architecture trials at a per-trial cost on the order of dollars rather than thousands of dollars, and at a daily throughput on the order of one hundred runs rather than one. At that rate the experiment count crosses a threshold where empirical Bayes on g_A becomes meaningful: one can speak, with calibrated uncertainty, about the expected marginal value of the next experiment.

Once that threshold is crossed, architecture search is structurally identical to data acquisition. Both consume capital. Both produce experiments. Both yield distributions of outcomes whose mean shifts as the campaign continues. The only structural difference is the channel through which the value is delivered: data acquisition makes more of the world available to the model; architecture search makes the model more efficient per unit of world available.

Empirical evidence from an 11-day campaign

We ran an autonomous architecture-search campaign in which a single human researcher operated the harness, validated each candidate architecture’s performance gain by hand before the running-best pointer was advanced, and was compensated as carry on the realized improvement in annual returns plus a small base salary.

Campaign summary.

Over 11 calendar days the harness produced 929 candidate architectures. Each was passed through a common backtest-and-validation loop. After cohort filtering (Appendix E in the recommended structure below), the comparable retained sample contains 356 experiments. The total token cost was approximately **\$300**; the compute platform was a single RTX 5090.

Headline result.

The running-best validated Calmar reached **2.05** on the sealed holdout window. Over the same window the SPY benchmark posted a Calmar of approximately 0.373; the best-discovered architecture therefore outperformed the benchmark by roughly **5.5×**. Annualized return on the best architecture was 30%, against 14% for SPY over the same window. Excess return improved from approximately 0.15 at the start of the campaign to approximately 0.30 by the end.

Figure 1 (carry forward existing chart). Risk-adjusted return (Calmar) versus cumulative experiment count across the 11-day campaign. Running best advances as a staircase. SPY benchmark (Calmar ≈ 0.37 over the same window) shown as horizontal reference.

Source figure in current draft: lib/diagrams/calmar_chart.png. No change to the figure itself; the body text below replaces the surrounding prose.

Distributional shape.

The running-best curve is a *staircase*, not a smooth monotone curve. Most experiments deliver no improvement; the tail does. The implication for pricing g_A is that the relevant statistic is not the mean of all experiments but the conditional mean of the upper tail times the rate at which the upper tail is sampled. We

give the formal estimator in Appendix E. The reduced fact sufficient for the body is that the campaign’s empirical $\hat{g}_A$ is materially greater than zero — the marginal dollar spent on this search loop returned, conditional on the cohort filters, roughly an order of magnitude more than its cost in deployable expected return.

Implied response law and parallel research

The campaign also lets us write down a candidate response law for architecture search, in the same spirit as the data-scaling law of Section 5.2. The natural axis is dollars spent on auto-research tokens (plus the implicit cost of the human carry); the natural response variable is the running-best validated risk-adjusted return on a fixed holdout.

We do not yet have enough samples to fit the exponent of this response law with the precision used for the data-scaling law. What we can say from the single-campaign evidence and from the structural argument is the following:

- **Single-loop concavity.** The response law is almost certainly concave in dollars spent on a single research loop, because the search distribution has a fat right tail and tail draws become rarer as the running best advances.
- **Parallel-loop additivity.** Adding *more* research loops (more researcher-harness pairs, run in parallel on disjoint regions of architecture space) shifts the response law up rather than along — each additional loop has its own running-best with its own concave growth, and the firm’s overall best is the maximum across loops. We expect the parallel-loop scaling itself to be a power law with diminishing returns from search-region overlap.
- **Lower-bound character of the result.** The throughput-per-dollar reported here is a lower bound on what is achievable. The campaign was conducted with relatively small base models on a single consumer GPU; performance results from this regime are conventionally a lower bound on what scaled architectures discover, because configurations that survive at small scale survive disproportionately at large scale.

These three observations together justify treating g_A in the working controller as bounded below by the campaign-derived estimate, with upside coming primarily from parallelization rather than from increasing the depth of any individual loop.

Methodology and selection-bias discipline

Two methodological points deserve to be made explicit, because architecture-search results are particularly vulnerable to overstatement.

Holdout discipline.

Every architecture in the campaign was evaluated on the same train / test / holdout split used by the deployment row Q — training on [2003, 2020), testing on [2020, 2025), reserving the post-2025 window as sealed holdout. The Calmar 2.05 figure is computed on the sealed holdout window. No architecture was selected on holdout performance.

Search-vs-test ratio.

The classical concern is that a sufficiently large architecture search will eventually find a configuration that performs well on the test window by chance. The defense is the standard one of López de Prado: as long as the number of distinct architectures evaluated is several orders of magnitude smaller than the number of independent samples in the test window, the false-discovery rate is bounded. The campaign comprised approximately 10^3 distinct architectures; the test window contains on the order of 10^7 hourly bars across the universe. The ratio is well within the safe regime.

The cohort-filtering rules (the path from 929 raw runs to 356 retained) and the explicit estimator for $\hat{g}_A$ appear in Appendix E.

Cross-row multiplicativity

The strongest reason architecture search deserves first-class treatment in the controller is not that it improves the deployment row in isolation. It is that an improved architecture changes the *front coefficient* of every other row's response law:

- A better architecture lowers $\kappa(A_t)$ in the data-scaling law, which means each marginal dollar spent on data acquisition (the X row) buys more loss reduction.
- A better architecture typically lowers the warm-start retraining cost c_{train}^{warm} for the same level of recovery, which raises g_R and shortens T_{loop} .
- A better architecture, by virtue of being more sample-efficient, often supports a wider deployable universe at the same risk budget, which raises the achievable scale of g_Q .

In the language of Section 6, this is the formal statement

$$\partial^2 J / \partial a_t^A \partial a_t^k > 0 \text{ for } k \in \{Q, X, R\}.$$

Architecture is the row whose marginal product enters every other row's marginal product as a multiplier. This is the structural reason architecture search is the deepest-leverage channel — and the reason a corporation that buys architecture search cheaply on the margin has a moat that compounds across the entire controller, not only within its own row.

What carries forward to the conclusion

The reframing above produces three ideas that belong in the conclusion of the paper rather than only in this section. We list them in the order they should appear there.

Suggested additions to the Conclusion

- **Architecture is the top-most row of the controller.** When the per-experiment cost of architecture search crosses below the variance of a single experiment's outcome, architecture stops being R&D overhead and becomes the term that determines the rate at which every other term improves.
- **Capital, not headcount, sets the pace.** Additional capital can be turned into additional research loops in parallel rather than into additional researchers competing for the same idle compute. This is what makes the moat compounding rather than additive.
- **Auto-research is a phase transition, not a productivity gain.** A 100× reduction in per-experiment cost is not just a faster research process — it is the difference between architecture research being narrative and architecture research being a priced asset class with measurable scaling laws.

Part II Recommended Appendix Structure

The current appendix carries roughly 30% literal duplication between Appendices A and B (the entire “Decision-Theoretic Implementation of the Corporate Loop” subsection appears verbatim twice; the “Local Linearized Allocator,” “Row Posteriors,” and “Direct Retraining Value vs. Possible Variance Reduction” blocks each appear twice as well). It also lacks an empirical appendix to back the architecture-search section above. The recommendation below removes the duplication, merges related material, adds the missing empirical appendix, and leaves the body unchanged.

The proposed structure is eight appendices. Three are merged from existing material, four are slim retitlings of existing appendices, and one (Appendix E) is new.

Proposed structure at a glance

#	Title	Source	Status
A	Formal Foundations and Decision-Theoretic Implementation	current A + dedupe of B	Merge
B	Channel Pricing and the Working Jacobian	current B (deduped)	Slim
C	Realized-Dollar Map and Market Frictions	current C	Slim
D	Data Scaling: Methodology, Fits, and Cross-Asset Tiers	current D	Keep
E	Architecture Search: Cohort Construction and the $\hat{g}_A$ Estimator	new	Add
F	Retraining Cost, Triggering, and Continual Learning	current B retraining + body §5.5	Merge
G	Accounting Bridge	current E	Slim
H	Notation Concordance	current F	Keep

Per-appendix detail and rationale

Appendix A. Formal Foundations and Decision-Theoretic Implementation — merge of current A with the redundant subsection in current B

Contents:

- World-firm joint dynamics; small-firm factorization; regimes of firm-world coupling.
- Improvement under model uncertainty: the no-certainty-under-observational-equivalence proposition.
- Decision-theoretic implementation of the corporate loop: actions, flows, and stocks (a_t^* , $Y_{t \rightarrow t+1}^s$, K_t^{dep}); observe $\rightarrow$ belief; world-model rollout and act; simulated vs. realized transition.

Notes:

- Cut the entire “Decision-Theoretic Implementation” subsection from current Appendix B. It is identical to the one in current Appendix A.
- This is the only formalism appendix the reader needs in order to read everything else.

Appendix B. Channel Pricing and the Working Jacobian — current B, deduped

Contents:

- Conceptual vs. operational action vectors and the coordinate map Ψ_t .
- Local linearized allocator and the chain-rule reduction $\nabla_{a^{\text{op}}} J = (\partial \Psi / \partial a^{\text{op}})^T \nabla_a J$.
- Row histories $\mathcal{H}_t^x$, posterior moments $(\hat{g}_{r,t}, \sigma_{r,t}^2)$, and the upper-confidence sampling rule.
- Held-out validation and the $d_r \ll n_{r,t}^{\text{fit}}$ admissibility criterion.
- Uncertainty reduction vs. drift: the local variance balance and the $(n / \tau^2) \sigma^4 > q \Delta t$ condition.

Notes:

- Remove the duplicate “Decision-Theoretic Implementation” subsection — it now lives only in Appendix A.
- Move the retraining-cost and trigger material out of this appendix and into the new Appendix F.

Appendix C. Realized-Dollar Map and Market Frictions — current C, slim**Contents:**

- Mechanism foundations: the market as a message-allocation-transfer object; auctions and continuous double auctions.
- Realized-dollar equation $Y_{t \rightarrow t+1}^{\$} = \sum \Delta p \cdot q - \Phi$.
- Friction decomposition $\Phi = \phi^{\text{impact}} + \phi^{\text{fees}} + \phi^{\text{fin}} + \phi^{\text{adv}}$; the square-root impact law.
- Message-to-cash timing: the institutional stages compressed into the body-level realized-dollar map.

Notes:

- Cut the “Simulation versus Realization” subsection at the end of current Appendix C — it is a third copy of the same material now in Appendix A.
- Cut the trailing duplicates of “Local Linearized Allocator” and “Row Posteriors and Admission” that currently sit at the end of Appendix C; they belong in Appendix B.

Appendix D. Data Scaling: Methodology, Fits, and Cross-Asset Tiers — current D, structurally unchanged**Contents:**

- Experimental setup: 36 runs across 12 universe sizes; fixed 56M-parameter architecture; train/test/holdout split.
- Three axes (tokens, dollar volume, geometric mean) and the argument for the geometric axis as the primary extrapolation axis.
- Two fit families (OLS log-axis vs. Kaplan power law) and the Taylor argument that justifies OLS for portfolio metrics.
- Functional-form robustness, leave-one-universe-out jackknife, and sensitivity to leverage points.
- Cross-asset universe table and tier construction with the quality-factor q methodology.

Notes:

- This is the strongest empirical appendix. Keep all substantive content. Suggested copy-edit only: prune the parenthetical engineering placeholders about the token-count convention into a single endnote so the running text is uninterrupted.

Appendix E. Architecture Search: Cohort Construction and the $\hat{g}_a$ Estimator — **new** — required to support revised §5.3

Contents:

- Raw artefact: 929 distinct experiments; 736 normalized identifiers.
- Cohort filter chain: removal of hard-spurious runs, failed / pending runs, runs without comparable primary metrics. Retained comparable cohort: 356 experiments.
- Metric definitions: Calmar / Sortino / annualized return on the sealed holdout window; relationship to the same window used by the Q row.
- Selection-bias bound: search-cardinality vs. independent-sample count in the test window; the López-de-Prado-style false-discovery argument used in the body.
- Tail-conditional estimator for $\hat{g}_A$: $\hat{g}_A \approx \pi_{\text{tail}} \cdot E[\Delta J \mid \text{tail}] - \text{cost-per-experiment}$, with bootstrap confidence intervals from the 356-cohort.
- Single-loop concavity vs. parallel-loop additivity: forecast form for budget allocation across multiple researcher-harness pairs.

Notes:

- This is the only new appendix required by the rewritten body. Without it, the “ $\hat{g}_A$ is materially greater than zero” claim in the body is unbacked.
- Recommended length: 3–4 pages. The estimator and selection-bias material are the substantive portion; the filter chain can be a single paragraph plus a small table.

Appendix F. Retraining Cost, Triggering, and Continual Learning — merge: current B retraining material + body §5.5 figure

Contents:

- Cost scaling: $c_{\text{train}}^{\text{cold}}$ and the warm-start discount p .
- Optimal-stopping trigger: cumulative wait loss vs. retraining cost plus uncertainty penalty.
- Direct retraining value $g_{\text{train}}(t)$ and the indicator-based formulation.
- Single-epoch convergence evidence: Theil-Sen Kaplan fits for first-epoch and best-epoch loss; intersection at $\approx 10^{12}$ dollar-weighted tokens.
- Cross-row variance reduction as an empirical extension, not a body-level assumption.

Notes:

- The continual-learning figure currently in the body of §5.5 should move here with its methodology paragraph.
- The body should retain only the conclusion that single-epoch training is feasible at current scale; the curve fit and its R^2 diagnostics belong in this appendix.

Appendix G. Accounting Bridge — current E, slim

Contents:

- State blocks: assets, liabilities, equity; the accounting identity.
- Timing-consistent bridge equations from realized dollars to next-period deployable capital.
- Reserves “ Res_t ” as the non-redeployable liquidity cushion.

Notes:

- Keep this appendix short. Its job is auditable plumbing — one page is enough for any reader who wants to verify that the controller’s deployable-capital object maps cleanly to a balance sheet.

Appendix H. Notation Concordance — current F, unchanged

Contents:

- Paper notation $\leftrightarrow$ implementation/system notation table.
- Reduced corporation tuple, world model, committed action, realized dollar flow, deployable capital, operational controls, working Jacobian.

Notes:

- Useful as the last appendix. No changes recommended.

Material to cut

The following passages are duplicates and should be removed:

- The entire “Decision-Theoretic Implementation of the Corporate Loop” subsection in current Appendix B (identical to the one in Appendix A).
- The duplicate “Local Linearized Allocator,” “Row Posteriors and Admission,” and “Direct Retraining Value vs. Possible Variance Reduction” blocks at the end of current Appendix C.
- The “Simulation versus Realization” subsection at the end of current Appendix C (third copy of the same material).

Net effect: the appendix shrinks by approximately three pages of pure duplication and grows by approximately three pages of new architecture-search methodology, leaving overall length roughly unchanged but raising signal-per-page substantially.

Editorial summary

The two changes proposed here are coupled. The body section is rewritten to argue that architecture search is a priced asset class, with cross-row multiplicativity as the structural justification. That argument depends on the cohort-construction and estimator discipline that does not currently appear anywhere in the paper, which is what the new Appendix E supplies. The remaining appendix changes are housekeeping: removing the literal duplications between Appendices A, B, and C, and moving the retraining-cost material into a single dedicated appendix. After these edits, every empirical claim in the body has exactly one appendix to back it, and the appendix has no internal duplication.

End of editorial revision document.